The Design of Sampling Strata for the National Household Food Acquisition and Purchase Survey

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Abstract

The National Household Food Acquisition and Purchase Survey (FoodAPS), sponsored by the United States Department of Agriculture's (USDA) Economic Research Service (ERS) and Food and Nutrition Service (FNS), examines the food purchasing behavior of various subgroups of the U.S. population. These subgroups include participants in the Supplemental Nutrition Assistance Program (SNAP) and the Special Supplemental Nutrition Program for Women, Infants, and Children (WIC), as well as households who are eligible for but don't participate in these programs. Participants in these social protection programs constitute small proportions of the U.S. population; obtaining an adequate number of such participants in a survey would be challenging absent stratified sampling to target SNAP and WIC participating households. This document describes how the U.S. Census Bureau (which is planning to conduct future versions of the FoodAPS survey on behalf of USDA) created sampling strata to flag the FoodAPS targeted subpopulations using machine learning applications in linked survey and administrative data. We describe the data, modeling techniques, and how well the sampling flags target low-income households and households receiving WIC and SNAP benefits. We additionally situate these efforts in the nascent literature on the use of big data and machine learning for the improvement of survey efficiency.

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Introduction

The United States Department of Agriculture's (USDA) National Household Food Acquisition and Purchase Survey (FoodAPS) is a survey of U.S. households that has as its objective the collection of unique and comprehensive data about household food purchases and acquisitions. One important requirement of this survey is to compare food purchase behavior among key subgroups. These subgroups are households receiving Supplemental Nutrition Assistance Program (SNAP) benefits; households participating in the Special Supplemental Nutrition Program for Women, Infants, and Children (WIC); low income households participating in neither of these programs; and higher income households participating in neither of these programs.

A key challenge in collecting data that yield reliable statistics for these subgroups is that they are small. For example, in an average month in 2022, only 6.3 million people in the United States received WIC benefits (Kessler et al. 2024), compared to the overall U.S. population of 333 million in 2022 (U.S. Census Bureau 2022), so only 1.9 percent of the population. Getting an adequate number of WIC participants in a survey with no explicit WIC sampling stratum would be unlikely without a very high overall sample size. While commercial data vendors provide flags to help target households meeting criteria of survey interest, it is unclear what information on income and program participation these vendor flags reflect and whether they enable survey administrators to adequately target WIC and SNAP households.

Faced with these challenges, the Demographic Research Area in the Center for Economic Studies (CES-Demo) created sample frame flags that indicate likely WIC participating, SNAP participating, and low income nonparticipating households (in either WIC or SNAP) to enable a stratified sample design that oversamples households in these subgroups. This report describes the methods that produce the sample frame flags for the FoodAPS Pilot, which is being conducted in 2025. These frame flags were developed by applying machine learning to linked survey, administrative, and third-party data.

Through various data sharing and acquisition agreements, the U.S. Census Bureau has acquired administrative data that can assist in efficiently sampling key populations. For example, the U.S. Census Bureau produces sampling strata for, and conducts, the National Survey of Children's Health (NSCH), a survey that targets households with children. The U.S. Census Bureau is able to produce reliable survey strata to improve sampling efficiency for the NSCH by drawing on high-quality administrative data that indicate the presence of children at an address and by supplementing the administrative data on child presence with a child-presence probability model estimated on linked administrative and American Community Survey data (U.S. Census Bureau 2020).

Recent research suggests sampling efficiency may be gained from using household level targeting models comparable to those employed for NSCH to identify survey populations of interest. Dutwin et al. (2024) compare survey frame flags developed using big data and machine learning that they term "Big Data Classifiers," or BDCs, to traditional sampling methods such as geographic targeting and commercial vendor flags. They find that BDCs offer better precision and recall than do geographic targeting and better precision than do vendor flags for a number of subpopulations of interest.

The goal of this work is to use administrative records, third-party data, and survey data to generate flags of SNAP and WIC participation at the household level as well as a flag for whether household income falls below 130 percent of the federal poverty level (FPL). These flags are appended to the sampling frame for FoodAPS to improve survey fielding efficiency. Our work focuses on identifying and integrating

data elements from various sources as model inputs, developing and testing predictive models for their out of sample prediction accuracy, and then applying the developed models to the most current data available so as to classify all units on the sample frame into survey strata. We produce frame flags that offer acceptably high levels of precision and recall in out-of-sample testing given the small size of the target populations, offering recall rates of 24 percent of the target population in the case of WIC, 56 percent in the case of SNAP, and 65 percent in the case of income below 130 percent FPL.[2]

We report on model development, evaluation, and application in the sections that follow. In the next section, we describe the data that serve as the input to the model.

Data

We link survey outcomes of income and program participation to a variety of administrative records (AR) and third-party data that are available for all units on the sample frame (the Census Bureau's address frame, or the Master Address File (MAF)) instead of merely survey respondents. Variables from these AR and third-party data serve as the predictive covariates in our models. As a result, this work requires the integration of many data sources. All data sources and their use are summarized in Appendix Table A1; they are described in detail in the subsections that follow.

Census Survey Data

Current Population Survey Annual Social and Economic Supplement (CPS ASEC): Indicators of household level WIC and SNAP participation and income below 130% of the federal poverty level – the left-hand-side, or target, variables in the models – are drawn from the CPS ASEC. Note that the CPS ASEC reference year (the year to which the data pertain) lag behind the CPS ASEC vintage by one year. This means that, e.g., the 2023 CPS ASEC data pertain to the calendar year 2022. See Table *3*. Data Vintages for details.

Census Planning Database: This is a harmonized dataset from the Census Bureau that largely consists of American Community Survey (ACS) 5-year estimates. We use tract-level estimates from this dataset as explanatory variables in our model. These data provide information on the percent of a geographic area that is in poverty or are children, for example. We supplement this information with household-level microdata derived from administrative data, which is discussed below. Using both geographic-level and household-level data on demographic and economic characteristics helps us greatly improve our model precision.

Administrative and Third-Party Data

In addition to the survey data used to produce the target variables and some of the features (*i.e.,* right-hand-side variables), we employ administrative and third-party data to produce additional features in model estimation and prediction. We detail these input data sources below. Where data were available

[2] In other words, among all households identified as "true" WIC participants based on survey data, our model successfully predicts 24 percent to be WIC participants. We thus interpret recall as a measure of coverage throughout this paper as it measures the percentage of the population that truly exhibits the characteristic to be oversampled that our predictions place in the relevant oversampled stratum.

at the person-level, we generated household level indicators (e.g., race/ethnicity of household head; an indicator for receipt of Medicaid by anyone in the household; etc.).

In the administrative and third-party data, not every observation with an address is successfully assigned a MAFID (the unique identifier in the MAF that is used as a linking key to merge datasets at the address level). Additionally, some administrative records have coverage gaps even for addresses successfully assigned a MAFID. For example, some low income households are not required by law to file tax returns, which would leave their household income missing in our tax data. If MAFID assignment or inclusion in administrative data for some other reason is nonrandom, models that rely on the data for input may yield biased estimates. To address this concern, we included an indicator for any household member being matched to any administrative or third-party data source as an explanatory variable. Additionally, for select administrative and third-party data sources, we included separate indicators for any household member being matched to each individual data source. Thus, we used missingness from these sources as an additional explanatory variable.

IRS Form 1040 tax return data: This is a tax-unit-level dataset, with person-level identifiers indicating which individuals belong to the tax unit. This dataset provides information on income (Adjusted Gross Income) and filing status as a proxy for marital status, among other observables, for all tax returns filed for a given tax year. These variables serve as determinants of SNAP and WIC eligibility and income falling below 130% of the federal poverty guideline.

Information Returns Master File (IRMF): This individual-level dataset provides information on every information return issued in the United States, for each of nine types of information returns:

1. Form W-2 (used to document income received from an employer),
2. Form 1099-INT (used to document interest income received),
3. Form 1099-DIV (used to document dividend income received),
4. Form 1099-R (used to document income received from a retirement account or pension),
5. Form 1099-SSA (used to document receipt of social security income),
6. Form 1099-MISC (used to document receipt of miscellaneous income, often income received for contracted services),
7. Form 1098 (used to document mortgage interest paid),
8. Form 1099-G (used to document income received from governments, such as state tax refunds or state unemployment insurance),
9. and Form 1099-S (used to document income received from sale of real estate).

For most of these forms, the IRMF only offers information on whether an individual received one of these forms, but not any information reported on the form. For these forms, the IRMF provides proxies for financial activity or asset holding. However, for Form W-2 and Form 1099-R, the IRMF also offers information on some of the income amounts and other details reported on the form. These income amounts are especially valuable to fill in data gaps for tax units that do not file 1040 tax returns.

Medicaid data from Centers for Medicare and Medicaid Services: This administrative dataset offers information on all individuals enrolled in Medicaid, an especially useful predictor for SNAP and WIC participation because of similarities in eligibility requirements across programs and the fact that Medicaid recipients are adjunctively eligible for WIC. According to the Survey of Income and Program Participation, 79.5 (79.4) percent of households that participated in SNAP (WIC) also participated in

Medicaid in 2022. However, Medicaid is a much larger program than SNAP or WIC. Among all households participating in Medicaid in 2022, only 39.4 percent (6.7 percent) also participated in SNAP (WIC). One major benefit of these Medicaid data is that this is a federal source that provides information for residents of every state + D.C.

United States Department of Housing and Urban Development (HUD) PIC-TRACS longitudinal file: This administrative dataset offers information on all households receiving government housing assistance, a useful predictor for SNAP and WIC participation because of similarities in eligibility requirements across programs. According to the Survey of Income and Program Participation, 19.9 (9.6) percent of households that participated in SNAP (WIC) also received rental subsidies in 2022.

Census Bureau's Master Address File-Auxiliary Reference File (MAF-ARF): The MAF-ARF contains information on who (if anyone) resides at each address on the Master Address File, based on IRS data and other administrative data sources. Using the MAF-ARF along with the HUD and Medicaid data improves our ability to identify people not present in the IRS data.

Housing data: We also link commercial housing data from Black Knight, a third-party vendor, and housing structure-type information from the sampling frame, the Master Address File (MAF). From the Black Knight data, we use information on the home's assessed value as well as whether it is owner-occupied. The home's value and whether the residence is a single-family home or in a multi-unit structure provide information on the socioeconomic status of households, particularly for households that we are unable to match to any other administrative data.

National Change of Address (NCOA) data from the U.S. Post Office (USPS): We use this source to produce features that offer insight into household level mobility (e.g., the number of times NCOA was filed for a given MAFID), so that we have some signal regarding how likely the household we observe in the administrative record data at that MAFID is to be the same one at the time of sampling.

Note that the FoodAPS Pilot did not have access to the Census Numident or data on all individuals who received benefits from the Social Security Administration (except for the indicators of 1099-SSA receipt coming from the Information Returns Master File). We are working to acquire access to these datasets for use in future versions of FoodAPS. While these datasets remain inaccessible, we pull demographic data from the 2010 Census. The 2010 Census provides comprehensive demographic data on individuals who were alive on April 1st, 2010. However, it does not provide demographic information for individuals who were born after April 1st, 2010. To fill this gap, we took extra steps for the FoodAPS Pilot to supplement 2010 Census data with age and sex data on recipients of Medicaid and housing assistance from the United States Department of Housing and Urban Development (HUD). While these datasets do not provide comprehensive coverage, they do provide age and sex observations for children in households that are especially likely to participate in SNAP or WIC or have income below 130 percent of the federal poverty guideline. For households that do not participate in Medicaid or housing assistance programs, we can identify the presence of children, if not their age or sex, based on dependent claiming among Form 1040 filers. We create an indicator variable that identifies when children of unknown age are present, and we include this variable on the right-hand side. While we are looking into the prospect of employing demographics from 2020 Census, this data source is not currently used to create sample frame flags for FoodAPS.

Additionally, SNAP and WIC state data were used to evaluate the performance of the frame flags, but the state data were not used in the development of the frame flag models. Our limited use of these data was due to the wording of the current agreements between the U.S. Census Bureau and state agencies, which allows for use of SNAP and WIC data indirectly for frame flag model improvement research, such as for model evaluation, but not directly as frame flag model inputs for production.

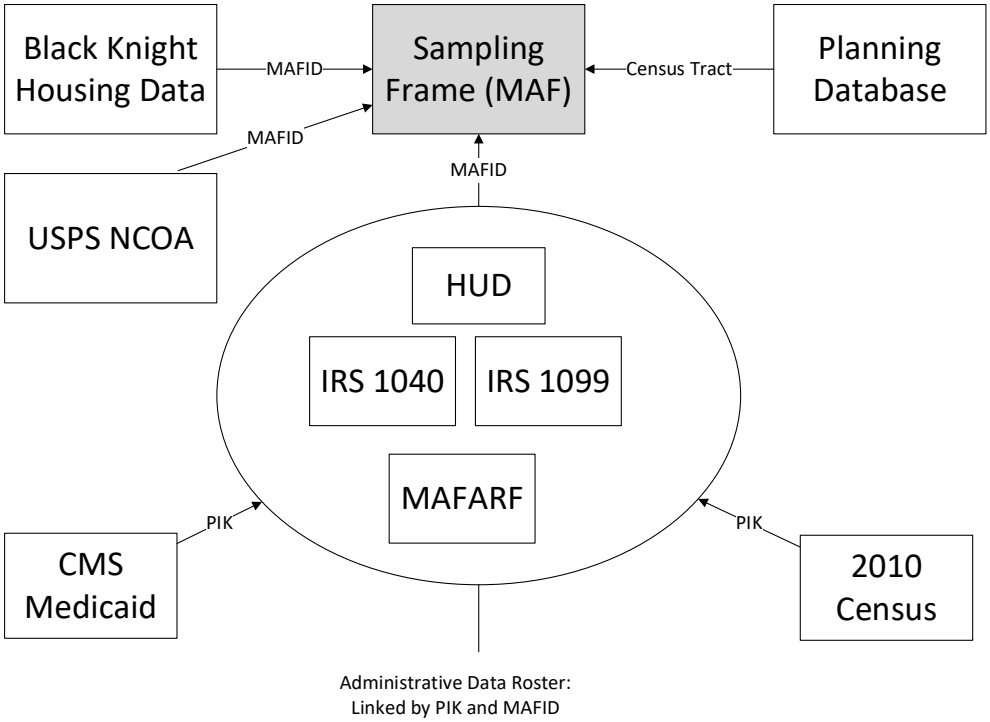
Linking

At the U.S. Census Bureau, datasets and data are processed through the Census Bureau's Person Identification Validation System (PVS), as described by Wagner & Layne (2014). This procedure matches both survey and administrative data to a master reference file. Individuals who are matched are given an identifier called a Protected Identification Key (PIK), which acts as an anonymized Social Security number that can be used to link administrative datasets and surveys. The linkage rates are very high for high quality administrative datasets that contain social security numbers, such as the IRS, HUD, and Medicaid data. Linkage rates are not quite as high for survey and decennial data, where name and date of birth, as well as sampled address are used to assign PIKs. In the 2010 Census, for example, about 90 percent of individuals were matched to a PIK (Wagner and Layne 2014). For more discussion on PVS and reasons an individual may not be assigned a PIK, see Bond et al. (2014).

Address identifiers share a common linking key, called a MAFID. For survey and decennial data, 100 percent of observations have a MAFID, since the MAF is used as a sampling frame for these sources. For the administrative data, MAFID assignment rates are high but incomplete. Looking at IRS 1040 data, Bee et al. (2015) found that about 90 percent of Form 1040 records matched to a MAFID, with unmatched tax records tending to be in the extreme upper and lower ends of the income distribution. If an address on the MAF is not matched to an IRS 1040 record, for example, there are two possible reasons for this. It could be that the household living at this address (if any) did not file their taxes. Or it could also be that they did file taxes but a problem with the address listed on their tax form resulted in the relevant MAFID not being assigned to the form. In our modeling, we are unable to separate these scenarios. The use of the Black Knight housing data and housing structure type from the sample frame should help provide some information on the socioeconomic status of such households.

Using these linkage keys, we combine the data in a manner depicted in Figure 1. Our first step is to link the IRS 1040, IRS 1099, HUD, and MAF-ARF data to a potential sampled address at the MAFID level, and then use these data to create a household roster. If someone's address conflicts between these sources, we give precedence to IRS 1040, IRS 1099, HUD, and MAF-ARF in that order. For example, for some people who move at the beginning of a year, their W-2 from their employer may be at their old address, but then in April they file taxes from their new address. We prioritize someone's IRS 1040 address over their IRS 1099 address for reasons like this. Once the household roster is created, we then link other sources on via PIK or MAFID.

Figure 1. Matching Diagram



Address Frame Extract

All these administrative data were linked together at the address level and matched to the MAF, the Census Bureau's address frame for household surveys. For the FoodAPS Pilot frame flags, we obtained the most up-to-date copy of the Master Address File from the Geography Division and assigned flags only to addresses with a valid postal address.

Administrative Data Strengths and Limitations

We end our data section by discussing the strengths and limitations the administrative data provide in predicting income and program participation, in order to provide intuition for the model performance results presented later. The administrative data to which we have access measures the following important sources of household level variation in program participation well:

- Income: From IRS data
- Number of household members: From list of people on IRS tax form and in the Medicaid and HUD Data
- Ages and sex of adults in household: From 2010 Census
- Presence of children in household: From IRS 1040 data, the list of dependents
- Program benefit receipt for Medicaid and HUD benefits, which are highly correlated with WIC and SNAP uptake.

Traditional sample designs oversample households exhibiting certain characteristics by randomly sampling units at higher rates in geographies where SNAP and WIC recipients are most densely located. We expect that using administrative and third-party microdata will improve our ability to target SNAP and WIC recipients relative to these more traditional sample designs that only target geographies, especially in geographies where the oversampled characteristic has a more heterogeneous distribution. Microdata can often provide more current information than area-level averages, so our approach also offers improvement over traditional area-based sampling designs in areas undergoing more change due to residential mobility or changes in residents' economic circumstances.

Measures related to WIC and SNAP receipt that our administrative data do not measure well are:

- Elements of categorical eligibility: We do not observe pregnancy and breastfeeding information that confers WIC eligibility and is therefore correlated with WIC receipt.
- Presence of children born after December 31st, 2023: The IRS data we use to create the final frame flags are from tax year 2023. Because a child had to be born by December 31st, 2023 in order to be claimed on that year's tax form, we do not observe any children born in 2024 or later.
- SNAP and WIC receipt: Records of SNAP and WIC participation from state data might serve as powerful features to predict the SNAP and WIC survey reports of participation that are our target variables. However, we have not used these data as such to date because our agreements do not currently support this use.

Therefore, while our data provide valuable information that should improve the efficiency of sampling households who receive SNAP or WIC benefits, our targeting will be imperfect given that we do not observe all relevant information.

Model Development

We train random forest models to estimate the probability of household level participation in WIC and/or SNAP, and household income below 130 percent of the FPL. We draw on CPS ASEC data for the target variables, and link AR data and tract-level survey estimates from the ACS via the planning database (PDB) for the feature set of predictive variables. In using survey data for the target variables and AR data for the feature set, we are able to train and validate models on the records that can be linked across the data sources, a subset of the full set of AR data. The parameterized models are then applied to the full set of AR, offering sampling frame flags for all in-scope units on the MAF.

Note that, while the feature set includes multiple data sources, including the third-party data and the PDB as detailed in the data section above, we gloss over these sources by referring to the input data to the feature set as the "AR data" throughout. For a detailed list of data sources drawn on for model development, see Appendix Table A1. We also refer to the linked data as the "CPS-AR data," again glossing over details in the interest of brevity. The training and testing data span the period 2017-2022. The data to which the trained models are applied to produce the FoodAPS Pilot frame flags is 2023 AR data, the most current AR data available at the time of the survey frame development.

Model Development Overview

Model development proceeded in several stages. First, we explored a number of different approaches to producing accurate predictions of program participation and FPL status. In this exploratory stage, we tried a variety of different algorithms including logit, extreme gradient boosting, neural nets, and random forest. We combined these algorithms with different minority oversampling techniques to address class imbalance given the small proportion of households participating in WIC and SNAP, including Synthetic Minority Oversampling Technique (SMOTE) (Chawla et al. 2002), Borderline SMOTE (Han et al. 2005), and Adaptive Synthetic Oversampling (ADASYN) (He et al. 2008). We also attempted different estimation strategies including independent estimation of each outcome and multiclass estimation.

Note that the FoodAPS Pilot, which will field in 2025, will occur three years after the most up to date training data, the final reference year of which is 2022, and two years after the most recent application data, AR data from 2023, are available.[3] In light of these time delays between model development, application, and survey fielding, we explored different timing relationships between features and targets in model development. We explored the performance of lagged features by lagging the features relative to the targets by three years, and we explored the performance of contemporaneous features by drawing features and targets from the same reference years. We evaluated the performance of the lagged models against testing data that was three time steps ahead, simulating what would happen in production if we estimated models with features from 2022 and targets from 2025 to predict income and program participation status in 2025. We also evaluated the performance of the contemporaneous models against testing data that was three time steps ahead, simulating what would happen in production if we estimated contemporaneous models on features and targets from 2022 to predict income and program participation status in 2025. The objective of each approach was to compare the apples-to-apples performance of the lagged and contemporaneous models in predicting outcomes under the time constraints presented by the gap between model development, application, and fielding.

Prior to exploring different estimation algorithms, oversampling techniques, independent and multiclass estimation, and different temporal relationships between features and targets, we divided the available data into training (80 percent of the linked data) and holdout (20 percent) sets; no MAFID could appear in both sets. The training set was used for model development, training, and hyperparameter tuning (via 3-fold cross validation), and all the exploratory testing of different estimation algorithms, oversampling techniques, independent versus multi-class estimation approaches, and contemporaneous versus lagged relationships between features and targets described above.

The holdout set was used for out-of-sample testing only once the exploratory stage of model development had been finalized – meaning, we tested the model in the holdout sample only once we had finished selecting the best performing estimation algorithm, oversampling technique, independent versus multiclass estimation, and temporal relationship between features and targets. We took this approach so that we could rely on the holdout set for an unbiased estimate of the final model's performance. Metrics by which model performance was evaluated and compared included precision, recall, F1 Score, model positive, and sample positive rates.

[3] Because Medicaid data are available at a two-year lag relative to the other administrative records and third-party data, the most recent Medicaid we can include in the AR that we apply to the full MAF is from 2021.

Random forest independent estimation without oversampling and with contemporary features and targets outperformed the other candidate approaches in the training data during our exploratory analysis; therefore, this approach was used to produce the final frame flags for the FoodAPS Pilot. We will not report further on our exploratory analyses here and will instead focus on the estimation and evaluation of the pilot model in the sections that follow.

Model Estimation

We first describe the general objectives of our model, then discuss the empirical process through which the final FoodAPS Pilot frame flag model was developed and tested.

Our objective is to estimate a binary flag indicating whether a household, conditional on a set of characteristics observed in our AR, will be expected to fall below 130 percent of the FPL, participate in SNAP, and/or participate in WIC. To achieve that goal, we first estimate the probabilities of those outcomes. We define $below130FPL_{ht}$ as a survey-based measure of whether the income of household *h* falls below 130 percent of the federal poverty guideline at time *t.*

With this CPS-derived relative income measure as the dependent, or target, variable, we can estimate the probability of household income falling below 130 percent of the federal poverty guidelines according to the general model shown in Equation 1 where: X_{ht} is a vector of observable household characteristics at time *t*, drawn from the AR data linked to the CPS ASEC; and θ is the vector of parameters to be estimated. The model is estimated on data from households, *h*, responding to the CPS ASEC.

$$P(below130FPL_{h\in CPS,t} = 1|X_{h\in CPS,t}) = f\left(X_{h\in CPS,t}; \theta\right) \quad \text{Equation 1}$$

The set of observable household characteristics used to estimate the model are those associated with the subset of households in the frame that can be linked to the CPS ASEC, $X_{h\in CPS,t} \subset X_{h\in MAF,t}$. The estimated parameters, $\hat{\theta}$, are applied to the full set of AR $X_{h\in MAF,t+1}$ to produce a probability estimate of whether income falls below 130 percent of the FPL for every household in our frame.

Note that, while the model is estimated contemporaneously, with both features and target drawn from time *t*, the model is applied to AR data from time *t+1*. This offset in time periods between the data on which the model is estimated and the data to which the estimated model is applied is employed to reflect the time constraints inherent in the data we have available for each step of the analysis and our objective of providing the most up-to-date estimates possible for the survey frame. More details on these time constraints are provided in the section on "*The Role of Time in Model Development and Application"* below.

$$\hat{P}(below130FPL_{h\in MAF,t+1}) = f\left(X_{h\in MAF,t+1}; \hat{\theta}\right) \quad \text{Equation 2}$$

We then discretize this continuous probability to obtain a binary indicator for whether the unit is predicted to have income below 130 percent of the FPL in year *t+1* by comparison to an empirically estimated threshold, $\hat{\rho}$. Predicted probabilities lying above this threshold are assigned a value of 1, indicating "below 130 percent of the FPL". Predicted probabilities lying below this threshold are assigned a value of 0, indicating "at or above 130 percent of the FPL".

However, an indicator for whether household *h* has income below 130 percent of the FPL in year *t+1* is not our ultimate target. Because the FoodAPS Pilot will be fielded two years after the latest reference year of AR available when the sample frame flags need to be finalized, our ultimate target is whether household *h* has income below 130 percent of the FPL in year *t+3*. To arrive at this target, we assume that, given the observable characteristics available at time *t+1*, the probabilities household *h* has income below 130 percent of the FPL in years *t+3* and *t+*1 have the same relationship relative to the estimated probability threshold $\hat{\rho}$:

$$I\left(\hat{P}\left(below130FPL_{h\in MAF,t+3}=1\middle|X_{h\in MAF,t+1}\right)>\hat{\rho}\right) = I\left(\hat{P}\left(below130FPL_{h\in MAF,t+1}=1|X_{h\in MAF,t+1}\right)>\hat{\rho}\right)$$

While this assumption is likely invalid in practice, this approach to predicting $I\left(\hat{P}\left(below130FPL_{h\in MAF,t+3}=1|X_{h\in MAF,t+1}\right)>\hat{\rho}\right)$ yielded better predictions than we obtained when taking a more direct approach by estimating models with features lagged relative to targets.

Similar to our approach to estimating $\hat{P}(below130FPL_{h\in MAF,t+1})$, we estimate the probability of a household participating in a social protection program following the general model shown in Equation 3 where: $participating_{p,h\in CPS,t}$ indicates whether household, *h*, responded to the CPS ASEC as participating in program, *p*={WIC, SNAP}, at time *t*; $X_{h\in CPS,t}$ is a vector of observable household characteristics at time *t* drawn from data available for all CPS ASEC households, including administrative records, third-party data, and Census-tract-level averages from the Planning Database; and ϕ_p is the vector of program specific parameters to be estimated.

$$P(participating_{p,h\in CPS,t}=1|X_{h\in CPS,t})=j\left(X_{h\in CPS,t};\phi_p\right) \qquad \text{Equation 3}$$

The estimated parameters, $\hat{\phi}_p$, are applied to the $X_{h\in MAF,t+1}$ of the households in the frame to produce a probability estimate of participation at time *t+1* for every in-scope household in the frame.

$$\hat{P}(participating_{p,h\in MAF,t+1})=j\left(X_{h\in MAF,t+1};\hat{\phi}_p\right) \qquad \text{Equation 4}$$

We then discretize this continuous probability to obtain a binary indicator for whether the unit is predicted to participate in program *p* in year *t+1* by comparison to an empirically estimated threshold,

$\widehat{\pi_p}$. Predicted probabilities lying above this threshold are assigned a value of 1, indicating "participating in program *p*". Predicted probabilities lying below this threshold are assigned a value of 0, indicating "not participating in program *p*".

As above, an indicator for whether household *h* participates in social protection program *p* in year *t+1* is not our ultimate target. Because the FoodAPS Pilot will be fielded two years after the latest reference year of AR available when the sample frame flags need to be finalized, our ultimate target is whether household *h* participates in social protection program *p* in year *t+3*. To arrive at this target, we assume that, given the observable characteristics available at time *t+1*, the probabilities household *h* participates in social protection program *p* in years *t+3* and *t+1* have the same relationship relative to the estimated probability threshold $\widehat{\pi_p}$:

$$I\left(\hat{P}\left(participating_{h\in MAF,t+3}=1\middle|X_{h\in MAF,t+1}\right)>\widehat{\pi_p}\right) = I\left(\hat{P}(participating_{h\in MAF,t+1}=1|X_{h\in MAF,t+1})>\widehat{\pi_p}\right)$$

While this assumption is likely invalid in practice, this approach to predicting $I\left(\hat{P}\left(participating_{h\in MAF,t+3}=1\middle|X_{h\in MAF,t+1}\right)>\widehat{\pi_p}\right)$ yielded better predictions than we obtained when taking a more direct approach by estimating models with features lagged relative to targets.

As noted above, after exploratory analysis that allowed us to compare the performance of a number of different algorithms, the contemporaneously estimated *independent random forest classifier without oversampling* stood out as the best performer across each of our targets. Consequently, the functions f() and j() above were implemented in Python using the *RandomForestClassifer* package for the Random Forest algorithm (Breiman, 2001) from scikit-learn (Pedregosa et al. 2011).

The outcomes of Equations 2 and 4, $\hat{P}(*_{h\in MAF,t+3})$ are probability estimates. To develop the survey strata, we use these probabilities to assign households to above/below 130 percent FPL classes as well as in WIC (SNAP)/not in WIC (SNAP) classes. In each case, we assign classes by identifying the F1 Score optimizing probability thresholds (i.e., $\hat{\rho}$ and $\widehat{\pi_p}$), where the F1 Score is the harmonic mean between model precision and model recall. Each household above the threshold is assigned to the "positive" class (meaning the household is in the particular subgroup of interest) and each household below the threshold is assigned to the "negative" class (the household is not in the subgroup of interest). Note that under the independent estimation of each target, the approach adopted for the pilot, a single household can be assigned to any combination of the three classes. The survey strata are then defined based on these class assignments, as shown in Table 1, such that the strata are exhaustive and mutually exclusive.

Table 1. Survey Stratum Assignment from Class Assignment

Stratum Number	Stratum Label	Stratum Assignment from Class Assignment
1	Household Receives WIC	*WIC* = 1 (regardless of predicted SNAP receipt)
2	Household Receives SNAP, No WIC	*SNAP* = 1 AND *WIC* = 0
3	Household Under 130 percent FPL, No Receipt of SNAP or WIC	*below130FPL* = 1 AND *SNAP* = 0 AND *WIC* = 0
4	Household Over 130 percent FPL, No Receipt of SNAP or WIC	*below130FPL* = 0 AND *SNAP* = 0 AND *WIC* = 0

Model Evaluation

Prior to estimating any models, we split the linked CPS-AR data into a training sample (80 percent of the total available linked data) and a holdout sample (20 percent), maintaining a strict separation between sets such that no MAFID could appear in both sets and no model exploration or training occurred in the holdout set.[4] We set aside the holdout set for an unbiased estimate of model performance. While strict separation between training and holdout data was maintained through the development and evaluation of the model, the final frame flags were produced on all available linked data.

Model performance is evaluated in the holdout sample using precision, recall, the F1 Score, and a comparison of the model positive rate with the sample positive rate. Each of these evaluation metrics are based on combinations of cells from the confusion matrix shown in Table 2; the confusion matrix tabulates the agreements (True Positives (TP) and True Negatives (TN)) and disagreements (False Negatives (FN) and False Positives (FP)) between the models' predicted outcomes and the true (i.e., survey-based) outcomes as observed in the holdout sample.

[4] We took care to divide the sample based on MAFID because the CPS ASEC has a longitudinal component. Dividing the observations into training and holdout samples without accounting for this longitudinal element risks placing one year of observations for a particular household in the training sample and the following year of observations for that same household in the holdout sample (or vice versa). This would potentially contaminate our evaluation of predictions based on models estimated in the training sample.

Table 2. Confusion Matrix

		True Outcome in Test Sample	
		Positive (1)	Negative (0)
Predicted Outcome in Test Sample	Positive (1)	True Positive (TP)	False Positive (FP)
	Negative (0)	False Negative (FN)	True Negative (TN)

From the cells in Table 2, we generate precision, a measure of survey fielding efficiency, as TP / (TP + FP);[5] recall, or a measure of coverage, as TP / (FN + TP); the model positive rate as (FP + TP) / (TN + FP + FN + TP), and the sample positive rate as (FN + TP) / (TN + FP + FN + TP). In addition to these evaluation metrics, we also report the F1 Score, the harmonic mean between precision and recall calculated as 2 * (precision * recall) / (precision + recall), for each model. Each of these evaluation metrics is summarized in Table 3.

Table 3. Evaluation Metrics

Metric	Formula
Precision	= TP / (TP + FP)
Recall	= TP / (FN + TP)
F1 Score	= 2 * (Precision * Recall) / (Precision + Recall)
Model positive	= (FP + TP) / (TN + FP + FN + TP)
Sample positive	= (FN + TP) / (TN + FP + FN + TP)

As noted above, we use the F1 Score to determine the probability threshold above which a household is categorized as positive or negative with respect to a given class. To avoid overfitting the threshold optimization routine on a single year of data, we determined the optimal probability threshold by averaging across a series of field year exercises, detailed below.

[5] Table 5 below reports that our WIC predictions have a precision value of 10.6 percent. In other words, among all households identified by our model predictions as WIC participants, 10.6 percent are "true" WIC participants based on survey data. We thus interpret precision as a measure of survey fielding efficiency throughout this paper as it measures the percentage of observations our model predicts to have a characteristic to be oversampled that truly exhibit that characteristic. FoodAPS oversamples units anticipated to contain WIC participants aiming to reduce the variance of WIC estimates. To the extent that predicted WIC-participating households are not "true" WIC-participating households, gaining survey responses from these households does not contribute to the goal of reducing the variance of WIC estimates.

Next, we address the complicated role of time in the development, evaluation, and application of the frame flag models.

The Role of Time in Model Development, Evaluation, and Application

We conduct different modeling simulation exercises on different vintages of data. Even within exercises, the vintages of the input data do not perfectly align due to data availability, since Medicaid data are available at a two year lag relative to the CPS ASEC reference year and the other AR data.

Table 4 displays the alignment of data vintages in the sample to which the trained model was applied to create the production sample frame flags (the first column, "AR Vintages Matched to the Entire MAF for Sampling") as well as some of the samples in which the model was developed and tested (the second and third columns); the same pattern applies to earlier CPS ASEC years not shown.

Table 4. Data Vintages

	AR Vintages Matched to Entire MAF for Sampling	AR Vintages for 2023 CPS ASEC Observations	AR Vintages for 2022 CPS ASEC Observations
CPS -ASEC reference year		*2022*	*2021*
IRS	2023*	2022*	2021*
HUD	2023	2022	2021
Medicaid	2021	2020	2019
Black Knight	2023	2022	2021
MAF-ARF	2023	2022	2021

* Indicates that we use the prior year's IRS 1040 data when the current year's data are unavailable. This can happen if people file their tax returns late or not at all.

The administrative and public data that we use for our feature set, and the income and program participation indicators drawn from survey data that serve as our target variables, are available on an annual basis. Consequently, we can only predict likely household relative income and program participation at an annual level. The FoodAPS Pilot will field three years after the most up-to-date training data, and two years after the most up-to-date application data are available (where the difference between "training data" and "application data" is that the training data are AR data linked to the CPS and the application data does not require CPS linkage as, once the model is parameterized, target data is not required). Specifically, at the time of model development (January-May 2024), the most current CPS ASEC data cover calendar year 2022, which means that we can train a model with targets from 2022. The most current AR data are from calendar year 2023, which means that we can apply the model, once estimated on data from 2022 and prior, to even more up to date AR data from 2023.

As described in the Model Estimation section above, an additional temporal challenge arises in that the model is developed in calendar year 2024 but the survey will not be in the field until March of 2025. To understand the performance of the frame flag models in the context of these temporal challenges, our evaluations of the model are conducted as "field year exercises" that simulate the time gaps between

training data inputs, application data inputs, and the additional years that will pass before the pilot reaches the field.

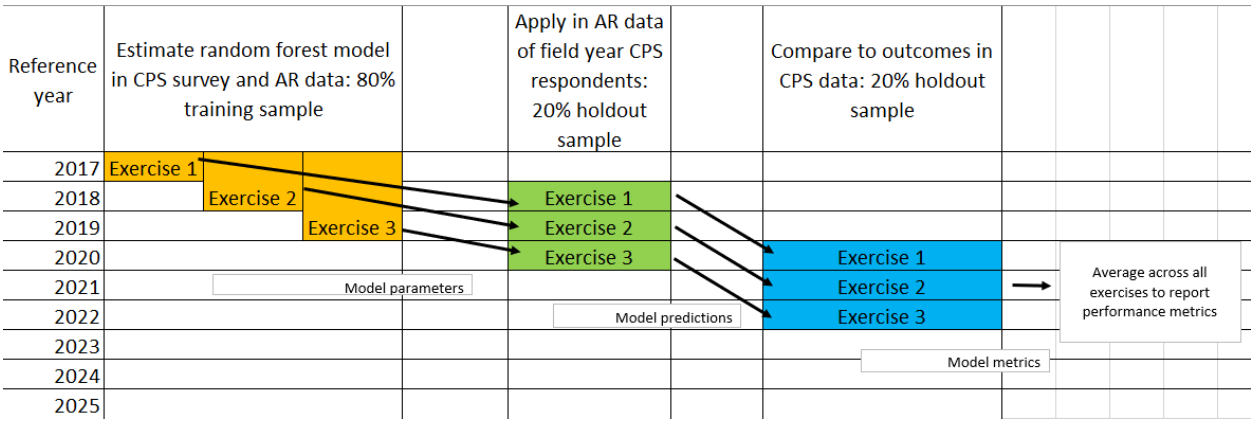
Evaluation in the Holdout Set

To account for these data lags, we perform model training and evaluation on data that were offset by two and three years across three field exercises as shown in Figure 2. A first exercise (follow the "Exercise 1" boxes across Figure 2) involved:

1. training the models on the training data that included income and program participation targets drawn from CPS ASEC survey data for reference year 2017 and AR-based features for 2017 for those CPS ASEC respondents (in other words, 80 percent of the available linked CPS-AR data from the 2018 vintage of the CPS ASEC; recall that the reference year and the vintage of the CPS ASEC differ by one year),
2. applying the model estimates to the 2018 AR-based features for the holdout sample of reference year 2020 CPS ASEC respondents (20 percent of the available linked CPS-AR data from the 2021 vintage of the CPS ASEC),
3. and then comparing the resulting class assignments to those same households' observed outcomes in reference year 2020 (as observed in the 2021 vintage of the CPS ASEC).

This exercise was repeated for Field Exercises 2 and 3 with the key differences between field exercises being differences in the training and testing years – equivalently the set of CPS ASEC respondents on which the model was trained and tested – and that the training data accumulated temporally as the field exercises progressed. The model evaluation metrics reported later in Table 5 are averaged across all three field exercises.

Figure 2. Field Year Exercises to Evaluate Model Performance in Holdout Sample

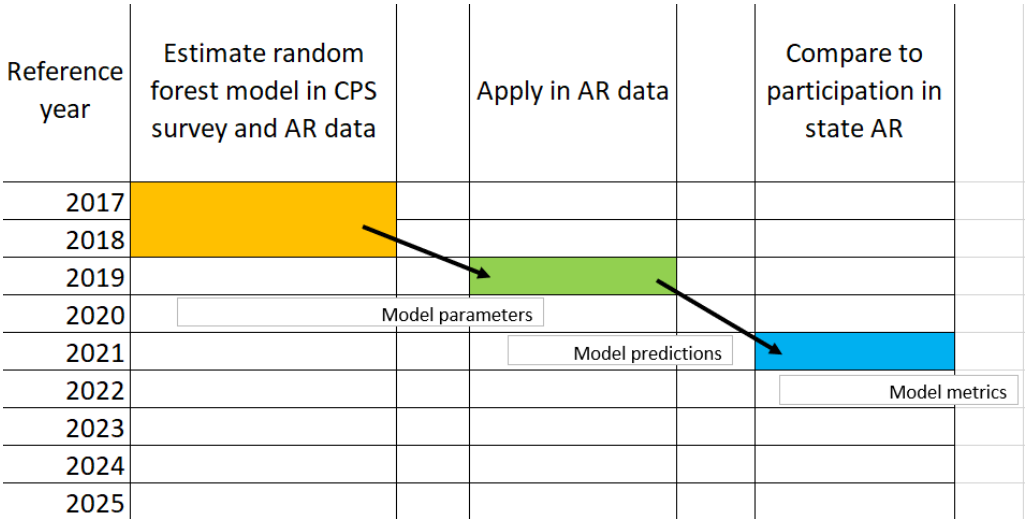


Evaluation in State SNAP and WIC Data

We additionally completed one field year exercise for reference year 2021, using state SNAP and WIC data as the outcome data against which model predictions were evaluated. As with the holdout data exercises above, this field exercise simulates the time gaps between model training, application, and the field year, and employs our best performing modeling approach. For this exercise, we trained the model on CPS-AR linked data from reference years 2017 and 2018. We used all available linked CPS-AR data to

train the model, reintegrating the training and holdout samples in the same manner as the final model for the pilot was produced. We applied the estimated model to 2019 AR data to predict the AR observations' classes, and then evaluated the class assignments against the AR observations' true participation outcomes in 2021, as recorded in the state WIC and SNAP data for that year. [6] We performed this analysis for the states that are at the intersection of two criteria: they are in the FoodAPS Pilot and they have given us permission to use their data for evaluations such as this. These include WIC data from Arizona, Kansas, Maine, and Montana and SNAP data from North Carolina and Nevada. For each of these states, the state program data provide a complete listing of every program participant over the course of the year, which makes these data superior out-of-sample data for this analysis. Note however, that we do not have data on both WIC and SNAP participation from any state, so while we can evaluate the accuracy of predicted SNAP participation, we cannot evaluate the accuracy of the stratum 2 – in SNAP, not in WIC – flag in this exercise. Note also that we do not have data on household income from the program data in this exercise so we cannot evaluate the accuracy of the below 130 percent of FPL class assignment.

Figure 3. Field Year Exercise to Evaluate Model Performance in State SNAP and WIC Data

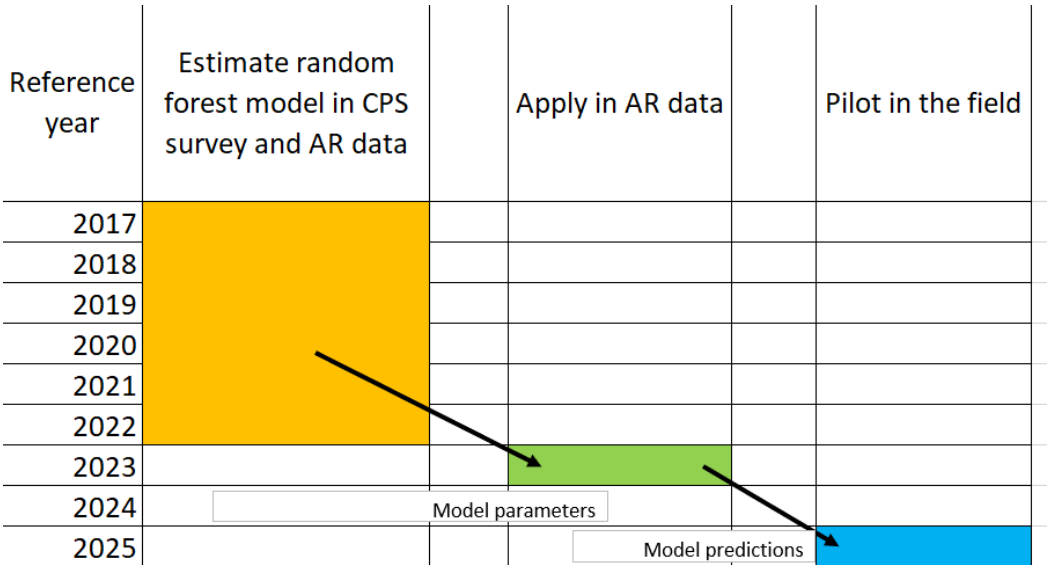


Estimation of Pilot Strata

The models that produce the sample frame flags for the pilot are trained on linked CPS-AR data from reference years 2017 – 2022; we train the models on all available years of data to avoid overfitting on the idiosyncrasies of any particular year and to increase the size of the target populations. The trained models are then applied to 2023 AR data for all in-scope units on the sample frame to produce the pilot frame flags. This process is diagramed in Figure 4.

[6] See Figure 3 for a summary of how this exercise is timed.

Figure 4. Estimation, Application, and Piloting



Model Evaluation Results

Table 5 displays model performance metrics in the hold out sample for each assigned class. The results shown are those averaged across the three field exercises detailed in Figure 2. Across these three field exercises, precision – the proportion of times the model is correct when it classifies a household into a given stratum – is 10.6 percent for WIC participants, 28.7 percent for SNAP participants and 37.9 percent for below 130 percent FPL households. Recall – the proportion of households truly in a given stratum that the model classifies as such – is 24.0 percent for WIC participants, 56.6 percent for SNAP participants and 65.4 percent for below 130 percent FPL households.

In comparing the model positive rate to the sample positive rate for each class in Table 5, it is clear that the models will produce false positives as they identify more than twice as many WIC participating households as are available in the sample population, slightly less than twice as many SNAP participating households as are available in the sample population, and 1.7 times more households below 130 percent FPL than are available in the sample population. A disadvantage of this tendency to produce oversampled classes that are too large is that it acts as a damper on the survey fielding efficiency of our approach. Nevertheless, our approach delivers strata within which the oversampled characteristic occurs more commonly than in a simple random sample (SRS). We can observe this increase in survey fielding efficiency by comparing the precision column to the sample positive rate column in Table 5: in the case of WIC, model precision is more than 5 times greater than the sample positive rate; in the case of SNAP, model precision is 2.5 times greater than the sample positive rate; and, in the case of 130 percent FPL, model precision is 1.8 times greater than the sample positive rate. These are significant improvements over SRS for the targeted subpopulations.

Table 5. Model performance in the CPS-AR holdout samples (averaged across the three field exercises)

	Precision	Recall	F1	Model Positive	Sample Positive
WIC	10.60	24.00	14.71	5.42	2.40
SNAP	28.70	56.60	38.09	22.29	11.31
Below 130 FPL	37.88	65.40	47.97	35.86	20.76

Source: Models estimated in, applied to, and evaluated against 2017-2022 CPS ASEC linked to administrative data; authors' estimations.

In Table 6, we cross-tabulate the assigned strata (based on class assignment from the model probability estimates, as detailed in Table 1 above) using the CPS-AR holdout sample field exercises against the observed strata in the CPS ASEC holdout sample. The assigned strata are as indicated in columns 1 and 2 while the observed strata in the CPS ASEC holdout samples are indicated in the columns along the top of the table. Note that the tabulations are reported such that the rows sum to 100, meaning we are looking at the distribution of the assigned strata across the observed strata. The table can be read as follows: of all the households assigned to, e.g., stratum 1 across the three field exercises, 17.05 percent are found in stratum 1 in the CPS ASEC holdout sample, 20.55 are found in stratum 2, 18.14 in stratum 3, and 44.26 in stratum 4. Because stratum 4 – the above 130% FPL, non-program-participation stratum – is the largest population stratum, a large proportion of the misassigned flags land there. Nevertheless, Table 6 shows that even when our model misassigns a unit to one of the oversampled classes, these misclassified units often belong to another oversampled class in truth, especially for the WIC and SNAP strata. For example, while only 17.05 percent of units assigned to the WIC stratum are observed to be WIC participants, more than half of these units are observed to be in one of the oversampled classes. While households misclassified as WIC participants would not contribute to a reduced variance of WIC estimates, they often contribute to a reduced variance for one of the other oversampled classes.

Table 6. Assigned Strata versus Observed Strata in the CPS-AR holdout sample (across the three field exercises)

		The Distribution of Households in Each Stratum observed in the CPS ASEC Holdout Sample				
Assigned Stratum Number	Assigned Stratum Label	Stratum 1	Stratum 2	Stratum 3	Stratum 4	Row Total
1	Household Receives WIC	17.05	20.55	18.14	44.26	100
2	Household Receives SNAP, No WIC	2.99	21.33	22.24	53.44	100
3	Household Under 130 FPL, No Receipt of SNAP or WIC	3.17	9.18	18.55	69.11	100
4	Household Over 130 FPL, No Receipt of SNAP or WIC	0.80	4.09	9.11	86.00	100

Source: Models estimated in, applied to, and evaluated against 2017-2022 CPS ASEC linked to administrative data; authors' estimations.

Table 7 displays model performance metrics in the 2021 state SNAP and WIC data following the field exercise detailed in Figure 3. In these data, we find precision of 17.4 percent in the case of WIC and 29.8 percent in the case of SNAP. Recall is 23.65 percent in the case of WIC and 66.90 percent in the case of SNAP. The stronger performance of the models as evaluated in the state SNAP and WIC data and shown in Table 7 relative to the performance of the models as evaluated in the CPS-ASEC hold out sample and shown in Table 5, may be due to the fact that program participation is often under-reported in surveys (Celhay et al. 2024). In other words, some classifications that are evaluated as false positives with reference to outcomes reported in a survey setting, may in fact be true positives when evaluated against state level records of participation. Additional analyses of the 2021 state SNAP and WIC field exercise are reported in Appendix Tables A2 and A3 where we report on the assigned strata.

Table 7. Model Performance in the 2021 State SNAP and WIC Data

	Precision	Recall	F1	Model Positive	Sample Positive
WIC	17.38	23.65	20.04	4.03	2.96
SNAP	29.80	66.90	41.23	32.35	14.41

Source: Model estimated in 2017-2022 CPS ASEC linked to administrative data; model applied to 2019 administrative records data; model predictions evaluated against 2021 state WIC and SNAP data; authors' estimations.

While machine learning combined with big data has been used to target low income households in other settings – notably, these methods have been used to develop models for the targeting of households for the efficient disbursement of social safety net benefits in developing economies (Altındağ et al. 2021; Banerjee et al. Forthcoming; McBride & Nichols 2018) – the use of predictive modeling to oversample demographic subgroups of interest is a new field in the context of survey frame development, with limited prior research available.

One of the demographic groups considered in Dutwin et al. (2024) largely overlaps with the below 130 FPL and SNAP populations targeted in this analysis: low income households, defined in Dutwin *et al.* (2024) as households with income below 25 thousand dollars per year ($25K). An annual income of $25K in 2021 is about 90 percent of the FPL for a family of four. In the 2022 ACS 5 year estimates (Table DP03 on data.census.gov), households with income below $25K/year (in 2022 inflation adjusted income) have a population proportion of 15.7 percent, which makes it a slightly larger target than our SNAP population target and a slightly smaller target than our 130 percent FPL target.[7] Although the size of these population targets differ, the performance metrics reported for the under $25K target in Dutwin et al. (2024) may offer context for those found in our study and offer insight into the extent to which household level targeting improves upon geographic targeting and vendor flags from commercial data providers. Dutwin et al. (2024) report metrics for the BDCs they have developed as well as for traditional oversample methods including geographic clustering and vendor flags; the F1 Scores for these approaches across the three out-of-sample exercises in Dutwin et al. (2024), denoted by source table and sample, are summarized in Figure 5. To this summary, we have added the F1 Scores for the SNAP and below 130 FPL models that we developed for FoodAPS and evaluated in out-of-sample exercises. We make this comparison as a means of offering rough, suggestive evidence that address-based sampling approaches can outperform sampling approaches based on area-level averages or modeled vendor-produced flags. It remains for future work to test this hypothesis in our context more directly.

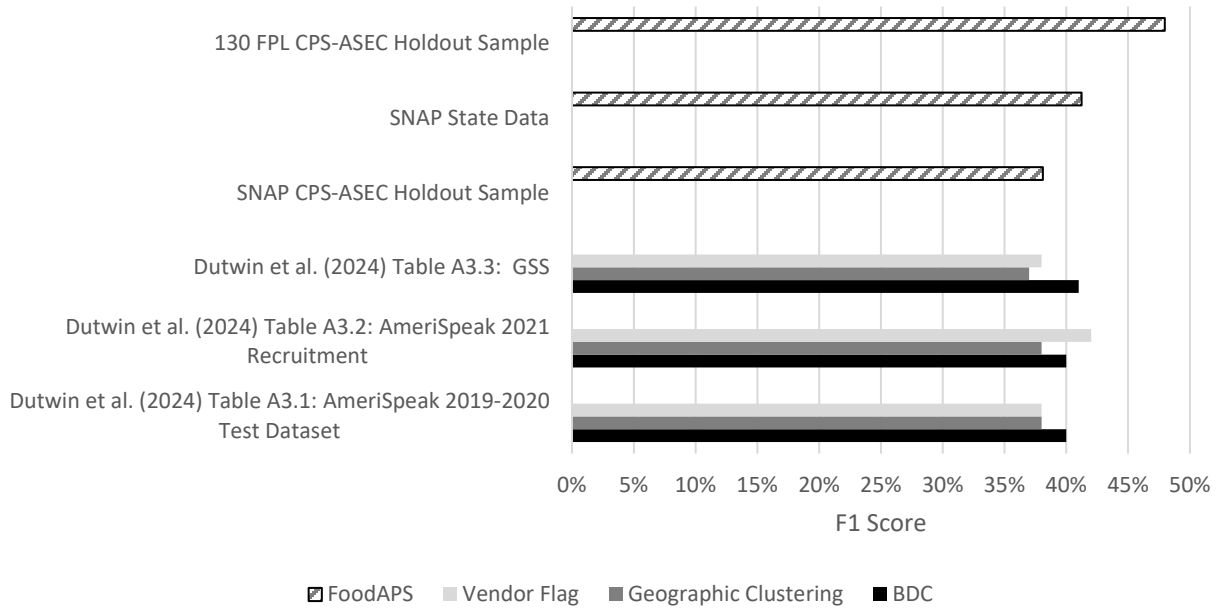
We first note in Figure 5 that the F1 Scores for the below 130 percent FPL and SNAP participating populations produced for the FoodAPS Pilot either exceed (in the case of 130 percent FPL target and the SNAP target evaluated in the SNAP state data) or land slightly below (the SNAP target evaluated in the

[7] We also note that the low income population target from Dutwin *et al.* (2024) is a much larger target than our WIC-participation target due to the categorical criteria restricting WIC eligibility beyond the income criteria; therefore we won't make comparisons between Dutwin *et al*. (2024) and the performance of the WIC model.

CPS-ASEC holdout sample) those produced by the $25K BDC developed in Dutwin et al. (2024). It should be noted that, while the F1 Scores are comparable, the Dutwin et al. (2024) BDCs offer higher precision while the FoodAPS Pilot frame flag models offer higher recall; these differences in precision and recall are largely due to differences in setting the class assignment probability threshold, so the F1 Score is the best point of comparison between the two. The slightly better F1 Scores for our 130 FPL target, and slightly worse for our SNAP target, relative to the Dutwin et al. (2024) $25K target are to be expected given the size of the $25K population compared to those of the FoodAPS targets.

The second thing to note in Figure 5 is that the performance of the Dutwin et al. (2024) BDCs and both the 130 FPL and SNAP FoodAPS Pilot models either match or exceed those of the geographic clustering approach and the vendor flags approach in all but one of the hold out samples (the vendor flags as evaluated in in the AmeriSpeak 2021 Recruitment sample outperform the BDC and our FoodAPS Pilot SNAP model as evaluated in the CPS-ASEC holdout sample) suggesting that predictive modeling can offer demonstrable gains in sampling efficiency and coverage over these traditional methods, at least insofar as the population target of low income households is concerned.

Figure 5. FoodAPS Pilot and Dutwin et al. (2024) F1 Scores in Holdout Samples



Source: Estimates from Dutwin et al. (2024) and from authors' estimates, as noted.

Finally, we report the assigned survey strata for the FoodAPS Pilot in Table 8. The strata are defined and reported for the states that will be included in the FoodAPS Pilot only; these states include Arizona, Kansas, Maine, Montana, West Virginia, Rhode Island, North Carolina, and Nevada. The FoodAPS Pilot strata are mutually exclusive and exhaustive with 4.84 percent of frame households assigned to the "WIC" stratum, 23.79 percent of frame households assigned to the "SNAP, no WIC" stratum, 9.16 percent of frame households assigned to the "under 130 FPL, no SNAP or WIC" stratum, and the remainder assigned to the "over 130 FPL, no SNAP or WIC" stratum.

Table 8. FoodAPS Pilot Survey Strata Distribution

Assigned Stratum	Stratum Label	Percent of MAF Valid Address Households in Pilot in each Stratum
1	Household Receives WIC	4.84
2	Household Receives SNAP, No WIC	23.79
3	Household Under 130 FPL, No Receipt of SNAP or WIC	9.16
4	Household Over 130 FPL, No Receipt of SNAP or WIC	62.21

Source: Model estimated in 2017-2022 CPS ASEC linked to administrative data; model applied to 2023 administrative records data; authors' estimations.

Conclusion

This report describes how linked survey and administrative data were used to produce sampling strata for the FoodAPS Pilot. A random forest classifier was used to estimate the likelihood of a household participating in WIC, SNAP, and/or falling below 130 percent of the FPL. These estimates were used to assign households to mutually exclusive and exhaustive sampling strata. While the strata assignments were challenging to estimate due to the low prevalence of program participants in the training data and underlying population, this approach to defining sampling strata improves considerably over simple random sampling and comparison with the nascent literature on the use of predictive modeling to define sampling strata suggests that it also outperforms geographic targeting and vendor flags in improving sampling efficiency.

The modeling strategy for future versions of FoodAPS may be refined as we obtain lessons learned from the fielding of the pilot and receive FoodAPS survey data that may help us improve on our model predictions. Finally, given the administrative data infrastructure that has been developed in pursuit of developing the FoodAPS Pilot frame flags, this work could be easily extended to other surveys to assist in the oversampling of key subpopulations.

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Appendix

Appendix Table A.1: Datasets used in Model Development

Agency/ Source	Dataset Name	Data Use
Survey and Decennial Data		
Census	Current Population Survey Annual Social and Economic Supplement	Dependent Variables-SNAP and WIC Receipt, Income < 130% FPL
Census	2010 Decennial Census	Explanatory Variables-Race and Ethnicity
Census	Planning Database (PDB)	Explanatory Variables-Geographic Characteristics
Administrative Data		
Census	Master Address File Auxiliary Reference File	Explanatory Variables-Household Composition
Census	Master Address File Extract	Explanatory Variables-Building Type
CMS	CMS Transformed Medicaid Statistical Information System (TMSIS)	Explanatory Variables-Age, Sex, Medicaid Receipt
Commercial	Commercial Black Knight Assessment Data	Explanatory Variables-Property Values, Renter Status
IRS	IRS Individual Master File (IMF 1040)	Explanatory Variables-Income and Household Composition
IRS	IRS Info. Returns Master File (IRMF W-2)	Explanatory Variables-Income
IRS	IRS Information Returns (1099-R) Master	Explanatory Variables-Income
IRS	IRS Information Returns Master File (IRMF 1099)	Explanatory Variables-Income and Household Composition
HUD	HUD PIC-TRACS Longitudinal File	Explanatory Variables-Age, Sex, Housing Assistance
USPS	National Change of Address (NCOA)	Explanatory Variables-Mobility
State records for North Carolina and Nevada	State - Supplemental Nutrition Assistance Program (SNAP)	Used only for out of sample testing
State records for Arizona, Kansas, Maine, and Montana	State - Women, Infants, and Children (WIC)	Used only for out of sample testing

Table A2. Survey strata compared with 2021 state SNAP and WIC Data: Of the WIC and SNAP participants we observe, which stratum are they placed in?

Assigned Stratum	Stratum Label	WIC Participant Proportion	SNAP Participant Proportion
1	Household Receives WIC	23.65	11.83
2	Household Receives SNAP, No WIC	39.81	55.39
3	Household Under 130 FPL, No Receipt of SNAP or WIC	9.52	8.53
4	Household Over 130 FPL, No Receipt of SNAP or WIC	27.01	24.25

Source: Models estimated in 2017-2022 CPS ASEC linked to administrative data; models applied to 2019 administrative records data; assigned strata from model predictions evaluated against 2021 state WIC and SNAP data; authors' estimations.

Table A3. Survey strata compared with 2021 state SNAP and WIC Data: What proportion of each assigned frame flag is composed on WIC and SNAP participants?

Assigned Stratum	Stratum Label	WIC Participant Proportion	SNAP Participant Proportion	WIC-AR Data Proportion	SNAP-AR Data Proportion
1	Household Receives WIC	17.38	41.28	4.03	4.13
2	Household Receives SNAP, No WIC	5.10	27.84	23.15	28.67
3	Household Under 130 FPL, No Receipt of SNAP or WIC	2.83	13.77	9.97	8.93
4	Household Over 130 FPL, No Receipt of SNAP or WIC	1.27	6.00	62.85	58.27

Source: Models estimated in 2017-2022 CPS ASEC linked to administrative data; models applied to 2019 administrative records data; assigned strata from model predictions evaluated against 2021 state WIC and SNAP data; authors' estimations.